



Hydragenix® TypHall Plus (AFCO 0238)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

Revision Date: 02/12/2024

Version: 1.0

SECTION 1: IDENTIFICATION

Product Identifier

Product Form: Mixture

Product Name: Hydragenix® TypHall Plus

Product Code: AFCO 0238

Intended Use of the Product

Use of the Substance/Mixture: Activator for the generation of acidified sodium chlorite and chlorine dioxide. For professional use only.

Name, Address, and Telephone of the Responsible Party

Company

AFCO

800 Development Avenue

Chambersburg, PA 17201

T: 800-345-1329

www.afcocare.com

Emergency Telephone Number

Emergency Number : 1-800-424-9300 (CHEMTREC)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the Substance or Mixture

Classification (GHS-US)

Met. Corr. 1 H290

Skin Irrit. 2 H315

Eye Irrit. 2A H319

Label Elements

GHS-US Labeling

Hazard Pictograms (GHS-US)



GHS05

GHS07

Signal Word (GHS-US)

: Warning.

Hazard Statements (GHS-US)

: H290 - May be corrosive to metals

H315 - Causes skin irritation

H319 - Causes serious eye irritation

Precautionary Statements (GHS-US)

: P234 - Keep only in original container.

P260 - Do not breathe vapors, mist, spray.

P264 - Wash hands thoroughly after handling.

P280 - Wear protective glove, protective clothing, eye protection, face protection.

P301+P317 - If swallowed: Get medical help.

P302+P352 - If on skin wash with plenty of soap and water.

P305+P351+P338 - If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P321 - Specific treatment (see section 4).

P332+P313 - If skin irritation occurs get medical advice/attention.

P337+P313 - If eye irritation occurs get medical advice/attention.

P362 - Take off contaminated clothing and wash before reuse.

P390 - Absorb spillage to prevent material damage.

P406 - Store in corrosive resistant container with a resistant inner liner.

Other Hazards

OSHA: Non-hazardous under OSHA Hazard Communication Standard.

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Unknown Acute Toxicity (GHS-US): Not available.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Substances

Mixture

Name	Product identifier	% (w/w)
Hydrogen sulfate - HSO_4^-	(CAS No) 14996-02-2	12.5 – 14.5
Hydronium - H_3O^+	(CAS No) 13968-08-6	3.5 – 4.5

Additional Information: None.

SECTION 4: FIRST AID MEASURES

Description of First Aid Measures

General: Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).

Inhalation: Not a likely route of exposure. Remove to fresh air if irritation occurs. If symptoms develop, obtain medical attention.

Skin Contact: Exposure to skin causes irritation or redness. No toxicity associated with the product being absorbed through skin. Wash exposed areas with water. In the unlikely event of irritation, seek medical advice.

Eye Contact: This product causes irritation to the eyes. Rinse with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Seek medical advice if irritation persists.

Ingestion: Ingesting large volumes may cause minor gastrointestinal distress due to separation of grease and oil in the digestive tract. Drink water to assist in digestion if swallowed. Seek medical advice.

Most Important Symptoms and Effects Both Acute and Delayed

Acute: Ingestion will cause minor distress to the gastrointestinal tract. May cause chemical and mechanical eye irritation with direct contact.

Indication of Any Immediate Medical Attention and Special Treatment Needed

Treat symptomatically. Seek medical assistance if required.

SECTION 5: FIRE-FIGHTING MEASURES

Extinguishing Media

Suitable Extinguishing Media: Use extinguishing media appropriate to surrounding fire conditions.

Unsuitable Extinguishing Media: None known.

Special Hazards Arising From the Substance or Mixture

Fire Hazard: Not flammable.

Explosion Hazard: Product is not explosive.

Reactivity: Containers may rupture from exposure to high temperatures, releasing contents that may be slippery. Chemically active on metals.

Advice for Firefighters

Precautionary Measures Fire: Exercise caution when fighting any chemical fire.

Firefighting Instructions: Extinguish preferably with dry chemical, foam or water spray.

Protection During Firefighting: Suitable protective clothing should be worn in fire conditions.

Hazardous Combustion Products: Sulfur oxides.

Reference to Other Sections

Refer to section 9 for flammability properties.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

General Measures: Avoid all contact with skin, eyes, or clothing. Avoid prolonged contact with metals.

For Non-Emergency Personnel

Protective Equipment: Use appropriate personal protection equipment (PPE).

Emergency Procedures: Evacuate unnecessary personnel.

For Emergency Personnel

Protective Equipment: Eye protection and face protection recommended.

Emergency Procedures: None.

Environmental Precautions

Avoid release to the environment.

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Methods and Material for Containment and Cleaning Up

Methods for Containment/Cleaning Up: Absorb spillage to avoid material damage. Rinse area with water. Dispose of material in accordance with local regulations.

Reference to Other Sections

See heading 7, 8, 13.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Additional Hazards When Processed: Avoid contact with eyes. Wear chemical resistant personal protective gear.

Hygiene Measures: Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work. Do not eat, drink or smoke when using this product. Wash hands and forearms thoroughly after handling.

Conditions for Safe Storage, Including Any Incompatibilities

Technical Measures: Comply with applicable regulations.

Storage Conditions: Store in closed containers between 35°F/1.6°C and 120°F/48.8°C.

Incompatible Materials: Avoid prolonged contact with metals.

Specific End Use(s): Activator for the generation of acidified sodium chlorite and chlorine dioxide. For professional use only.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

No Occupational Exposure Limits (OELs) have been established for this product or any of its chemical components.

Exposure Controls

Appropriate Engineering Controls: Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Ensure all national/local regulations are observed.

Personal Protective Equipment: Protective goggles. Protective clothing. Gloves. Face shield.



Materials for Protective Clothing: Chemically resistant materials and fabrics.

Hand Protection: Acid resistant gloves recommended.

Eye Protection: Safety glasses including face protection mandatory.

Skin and Body Protection: Wear suitable protective clothing.

Respiratory Protection: Usually not needed.

Engineering Controls: No special controls required.

General Hygiene Considerations: Handle according to established industrial hygiene and safety practices.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on Basic Physical and Chemical Properties

Physical State	: Liquid.
Appearance	: Clear.
Odor	: None to mild.
Odor Threshold	: Not available.
pH	: 0.1 - 1.2
Evaporation Rate Water =1	: 1
Melting Point	: N/A.
Freezing Point	: Not available.
Boiling Point	: >112°C (233.6°F)
Flash Point	: >220°C (428°F)
Auto-ignition Temperature	: Not flammable.
Decomposition Temperature	: Not available.
Flammability (solid, gas)	: Not flammable.
Lower Flammable Limit	: Not flammable.
Upper Flammable Limit	: Not flammable.
Vapor Pressure	: Not available.

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Relative Vapor Density at 20°C	: Not available.
Specific Gravity	: 1.065 – 1.085 g/mL
Solubility	: Soluble in water.
Partition coefficient: n-octanol/water	: Not available.
Viscosity	: Not available.
VOC Content, wt. %	: 0%
% Volatile, wt. %	: 0%
Sensitivity to Mechanical Impact	: None.
Sensitivity to Static Discharge	: None.

SECTION 10: STABILITY AND REACTIVITY

Reactivity: No additional information available.

Chemical Stability: Stable under normal conditions. Avoid temperature extremes.

Possibility of Hazardous Reactions: No hazardous reactions observed.

Conditions to Avoid: Do not freeze. Do not use above ambient temperature. Incompatible materials.

Incompatible Materials: Avoid prolonged contact with metals.

Hazardous Decomposition Products: Sulfur oxides.

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects - Product

Acute Toxicity: Not classified.

LD50 and LC50 Data: Not available.

Skin Corrosion/Irritation: Causes skin irritation.

Serious Eye Damage/Irritation: Contact causes serious eye irritation.

Respiratory or Skin Sensitization: Not classified.

Germ Cell Mutagenicity: Not available.

Teratogenicity: Not available.

Carcinogenicity: This product does not contain any carcinogens or potential carcinogens as listed by ACGIH, IARC, OSHA, or NTP.

Specific Target Organ Toxicity (Repeated Exposure): Expected to be similar to single exposures.

Reproductive Toxicity: Not available.

Specific Target Organ Toxicity (Single Exposure): Not classified.

Aspiration Hazard: Not classified.

Symptoms/Injuries After Inhalation: Product may be irritating to nasal tissue.

Symptoms/Injuries After Skin Contact: Prolonged contact may result in slight irritation.

Symptoms/Injuries After Eye Contact: Contact causes serious eye irritation.

Symptoms/Injuries After Ingestion: Product may be irritating to gastrointestinal tract.

Information on Toxicological Effects - Ingredient(s)

Other Information: None.

SECTION 12: ECOLOGICAL INFORMATION

Toxicity Not expected to be harmful to soil environments. Not expected to cause long lasting harmful effects to aquatic life.

Persistence and Degradability

Hydragenix® TypHall Plus (AFCO 0238)	
Persistence and Degradability	Not established

Bioaccumulative Potential

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Bioaccumulative Potential	Not available.

Mobility in Soil Not available.

Results of PBT and vPvB Assessment: Not available.

Additional Information on Eco-toxicity: The product does not add to the AOX-value of effluent water (DIN 38409).

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Treatment Methods: Disposal should be in accordance with local, state or national legislation. For disposal within the EC, the appropriate code according to the European Waste Catalogue (EWC) should be used. Containers must not be punctured or destroyed by burning, even when empty.

Additional Information: Neutralization recommended before disposal.

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SECTION 14: TRANSPORT INFORMATION

Note: Hydrogen sulfate is a key ingredient in Typhall Plus. In accordance with the DOT shipping regulations located in 49 CFR 172.101 Hazardous Materials Table, Hydrogen sulfate is not considered a proper shipping name for transportation. The Hazardous Materials Table states that Sulfuric acid will be used as the proper shipping name for Hydrogen sulfate during transportation. Therefore, the proper shipping name for Hydrogen sulfate is "UN2796, Sulfuric acid with not more than 51% acid, 8, II."

14.1 In Accordance with DOT

Proper Shipping Name : Sulfuric acid with not more than 51% acid
Hazard Class : 8
Identification Number : UN2796
Label Codes : 8
Packing Group : II
ERG Number : 157



14.2 In Accordance with IMDG

Proper Shipping Name : Sulfuric acid with not more than 51% acid
Hazard Class : 8
Identification Number : UN2796
Packing Group : II
Label Codes : 8
EmS-No. (Fire) : F-A
EmS-No. (Spillage) : S-B



14.3 In Accordance with IATA

Proper Shipping Name : Sulfuric acid with not more than 51% acid
Packing Group : II
Identification Number : UN2796
Hazard Class : 8
Label Codes : 8
ERG Code (IATA) : 8P



14.4 In Accordance with TDG

Proper Shipping Name : Sulfuric acid with not more than 51% acid
Packing Group : II
Hazard Class : 8
Identification Number : UN2796
Label Codes : 8



SECTION 15: REGULATORY INFORMATION

US Federal Regulations

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SARA Section 311/312 Hazard Classes	Health hazard - Skin corrosion or Irritation. Health hazard - Serious eye damage or eye irritation.

TSCA STATUS

All components of this product are listed, or excluded from listing, on the United States Environmental Protection Agency Toxic Substances Control Act (TSCA) inventory.

US State Regulations

California Proposition 65 - This product does not contain any substances known to the state of California to cause cancer, developmental and/or reproductive harm. Emergency management, secondary containment, and signs - For 8.1A 1,000 kg or 1,000 L

Canadian Regulations

No additional information available.

SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

Revision date : 02/12/2024

02/12/2024

EN (English US)

5/6

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Other Information : This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200.

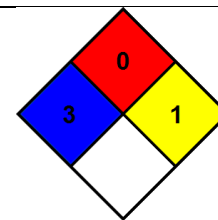
GHS Full Text Phrases:

Acute Tox. 5 (Oral)	Acute toxicity (oral) Category 5.
Met. Corr. 1	Corrosive to metals Category 1.
Eye Irrit. 2A	Serious eye damage/eye irritation Category 2A
Skin Irrit. 2	Skin corrosion/irritation Category 2.
H290	May be corrosive to metals.
H315	Causes skin irritation.
H319	Causes serious eye irritation.

NFPA Health Hazard : 3 - Short exposure could cause serious temporary or residual injury even though prompt medical attention was given.

NFPA Fire Hazard : 0 - Materials that will not burn.

NFPA Reactivity : 1 - Normally stable, but can become unstable at elevated temperatures and pressures or may react with water with some release of energy, but not violently.



HMIS III Rating

Health : 3 Serious Hazard - Major injury likely unless prompt action is taken and medical treatment is given.

Flammability : 0 - Minimal Hazard.

Physical : 1 - Slight Hazard.

Party Responsible for the Preparation of This Document

AFCO
800 Development Avenue
Chambersburg, PA 17201
T: 800-345-1329

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

North America GHS SDS 2015 (U.S., Can., Mex.)